



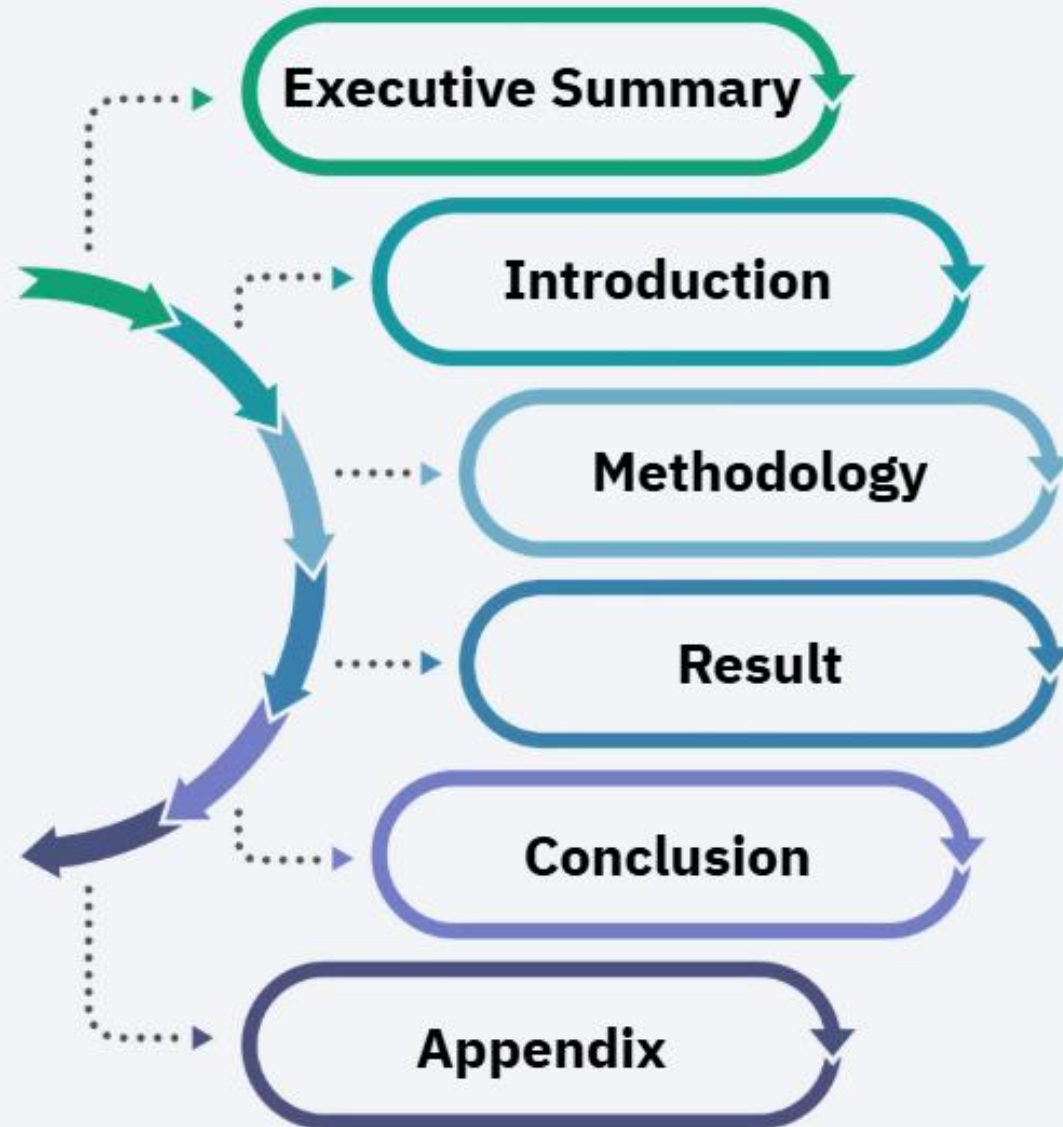
IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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2026/09/13

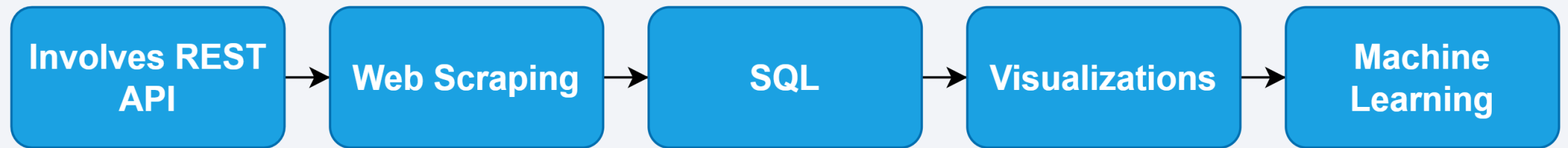


Outline



Executive Summary

Methodologies



The methodology outlines a sequential data science pipeline that begins with data collection via REST APIs and Web Scraping, followed by data management using SQL. The processed data is then explored through visualizations and ultimately used to train a machine learning classification model.

Result

The classification model is capable of predicting the landing success with an accuracy of 83.3%.

Introduction

Background

Commercial space launch providers traditionally discard the entire rocket after launch, costing upwards of \$165+ million per mission. SpaceX drastically reduces launch costs to ~\$62 million by recovering and reusing the Falcon 9 first stage.

Problem?

The cost savings are largely because SpaceX can reuse the first stage. Therefore, if we can determine if the first stage will land successfully, we can determine the actual cost of a launch.

Section 1

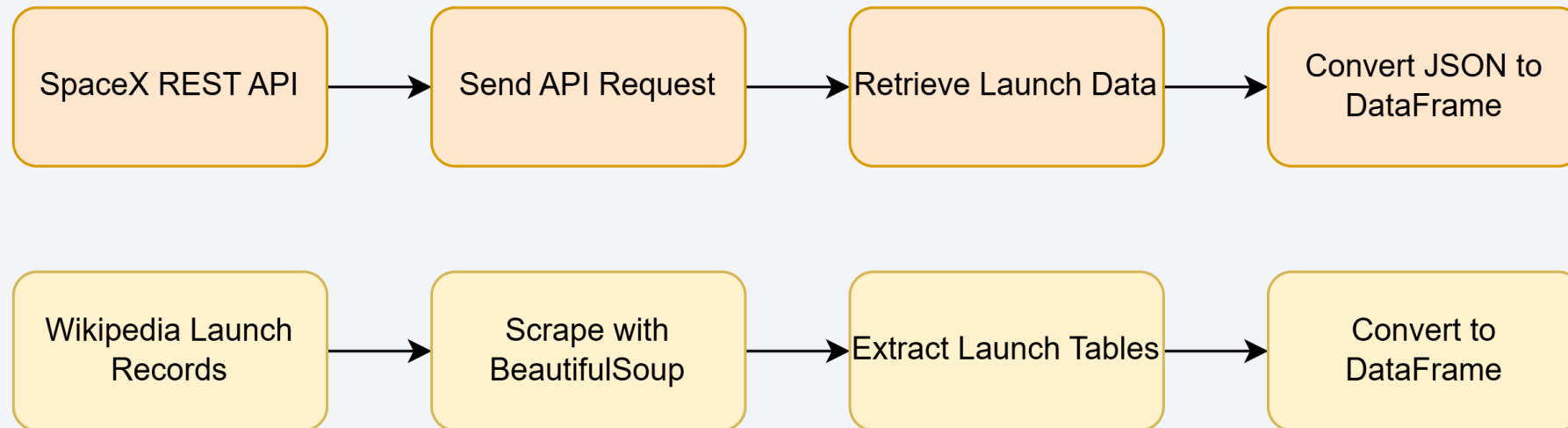
Methodology

Methodology

Executive Summary

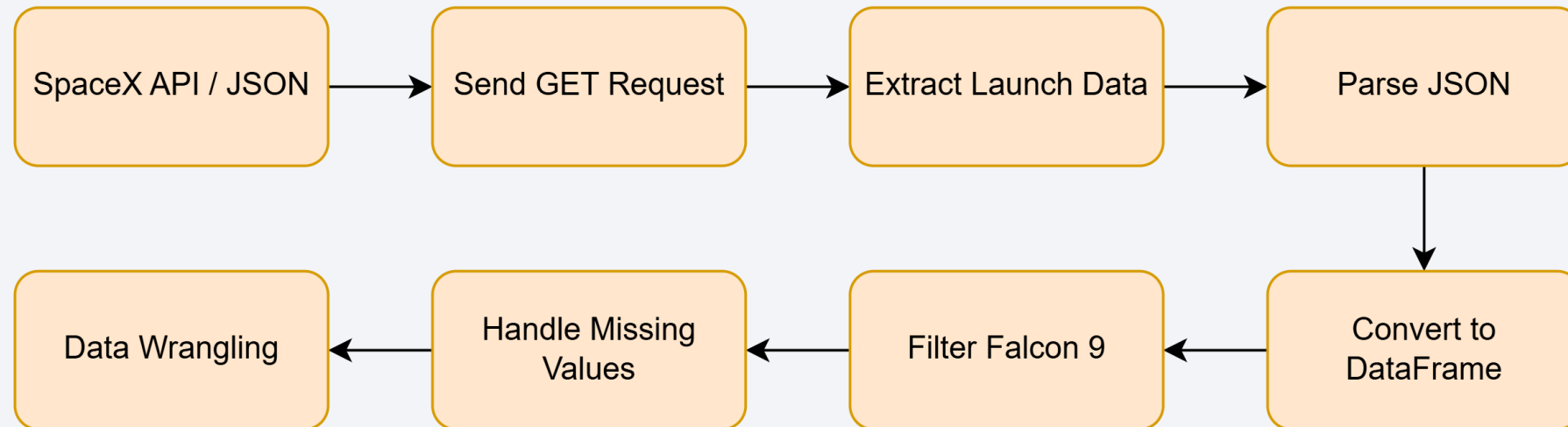
- Data collection methodology:
 - Collected data from the SpaceX API and web scraping from wiki pages.
- Perform data wrangling
 - Processed JSON objects and HTML tables, then transformed the data into Pandas DataFrames for further analysis and visualization.
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
 - Applied classification-based machine learning models to predict the successful landing of the Falcon 9 first stage.

Data Collection



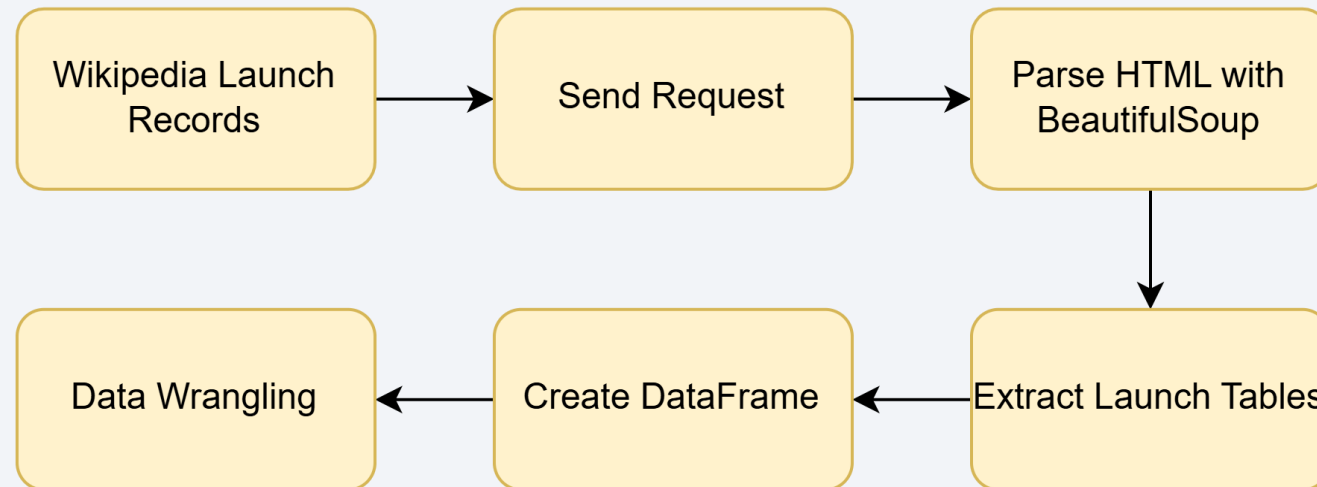
The historical flight data collection process was carried out from two sources: **SpaceX REST API** and **Wikipedia**.

Data Collection – SpaceX API



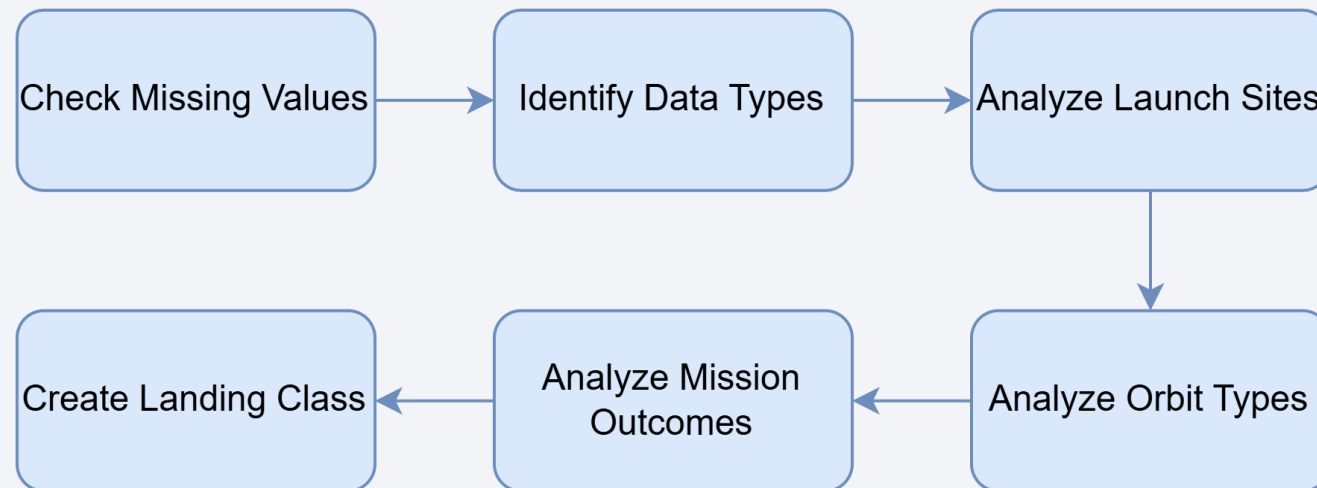
Data was collected from the **SpaceX API** using GET requests to retrieve historical launch data. The JSON response was then converted into a Pandas DataFrame for further processing.

Data Collection – Scraping



Data was collected from **Wikipedia** by sending a request to the launch records page and parsing the HTML content using BeautifulSoup. The extracted launch tables were then converted into a Pandas DataFrame for further data wrangling.

Data Wrangling



The data was analyzed by checking missing values, data types, launch sites, orbit types, and mission outcomes. The Outcome variable was then converted into a binary Class label, where 1 represents a successful landing and 0 represents an unsuccessful landing.

EDA with Data Visualization

Summary of Charts

- **Catplot** (Flight Number vs. Payload Mass) to compare payload and landing outcomes.
- **Catplot** (Flight Number vs. Launch Site) to examine landing outcomes across launch sites.
- **Catplot** (Payload Mass vs. Launch Site) to analyze payload distribution by launch site.
- **Bar Chart** (Orbit vs. Success Rate) to compare landing success across orbit types.
- **Catplot** (Flight Number vs. Orbit) to examine landing outcomes across orbit types.
- **Catplot** (Payload Mass vs. Orbit) to analyze payload and landing outcomes by orbit.
- **Line Chart** (Year vs. Success Rate) to observe the yearly trend in landing success.

EDA with SQL

Function of Queries

- **Launch Site** Analysis: Identify unique launch sites and sites beginning with “CCA”.
- **Payload** Analysis: Calculate total and average payload for customers and booster.
- **Landing** Analysis: Find the first successful ground-pad landing and analyze drone-ship results.
- **Mission Outcome** Analysis: Count successful and failed mission outcomes.
- **Maximum Payload** Analysis: Identify booster versions carrying the maximum payload.
- **Historical** Analysis: Examine 2015 failed drone-ship landing and rank outcomes by frequency.

Build an Interactive Map with Folium

Summary of Map Objects

- **Circle & Marker:** Mark launch sites and display their names on the map.
- **MarkerCluster:** Group launch markers to keep the map clear and readable.
- **MousePosition:** Identify coordinates of nearby points of interest.
- **PolyLine & Marker:** Show and measure distances between a launch site and nearby coastline, city, railway, and highway.

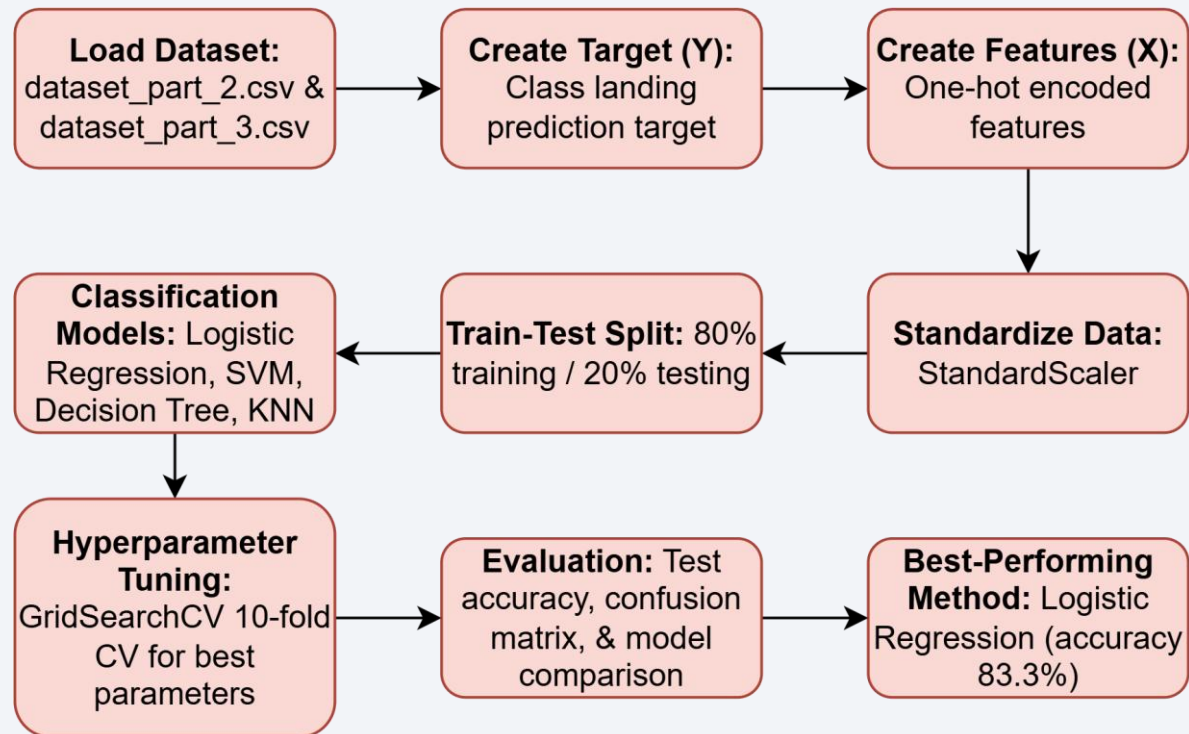
Build a Dashboard with Plotly Dash

The SpaceX Launch Dashboard was developed with interactive plots and controls to explore launch performance based on **launch site** and **payload mass**.

- **Launch Site Dropdown:** Allows users to select All Sites or a specific launch site. This makes it easier to compare launch performance across different sites.
- **Success/Failure Pie Chart:** Shows the number of successful and failed launches for the selected site, helping users quickly understand launch outcomes.
- **Payload Range Slider:** Filters launches by payload mass from 0–10,000 kg. This allows users to explore how launch outcomes vary across different payload ranges.
- **Payload Mass vs. Launch Outcome Scatter Plot:** Visualizes the relationship between payload mass and launch success/failure, with Booster Version Category used to distinguish the data points.
- **Interactive Callbacks:** The pie chart responds to the selected launch site, while the scatter plot responds to both the launch site and payload range, enabling users to explore different subsets of the launch data.

Predictive Analysis (Classification)

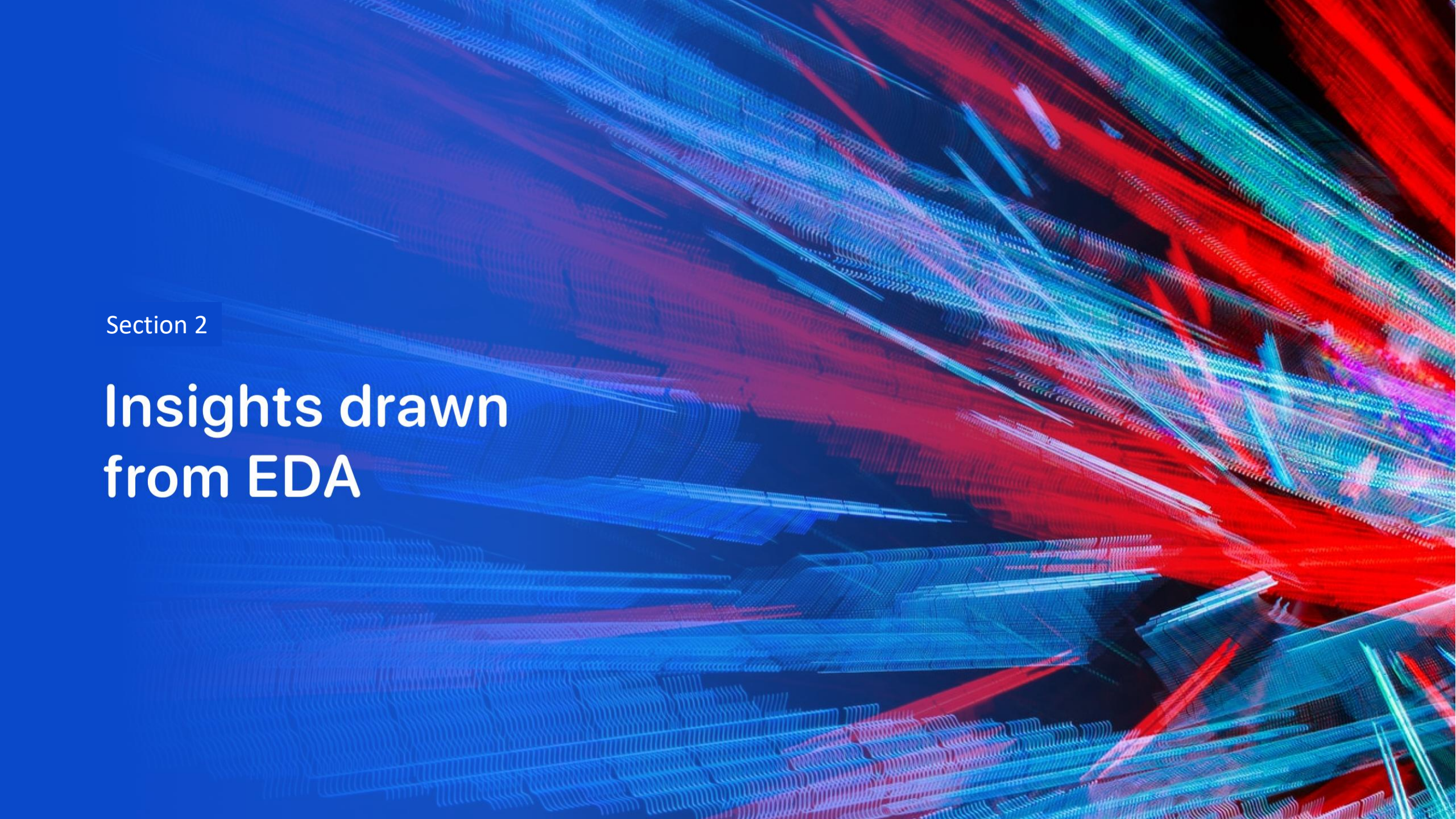
Model Development Process



- Data Preparation & Feature Engineering: Extracted key features and applied one-hot encoding to categorical variables.
- Feature Scaling & Split: Standardized features using StandardScaler and split data into 80% training and 20% testing.
- Model Training & Tuning: Evaluated four classification models and optimized hyperparameters using 10-fold GridSearchCV.

Results

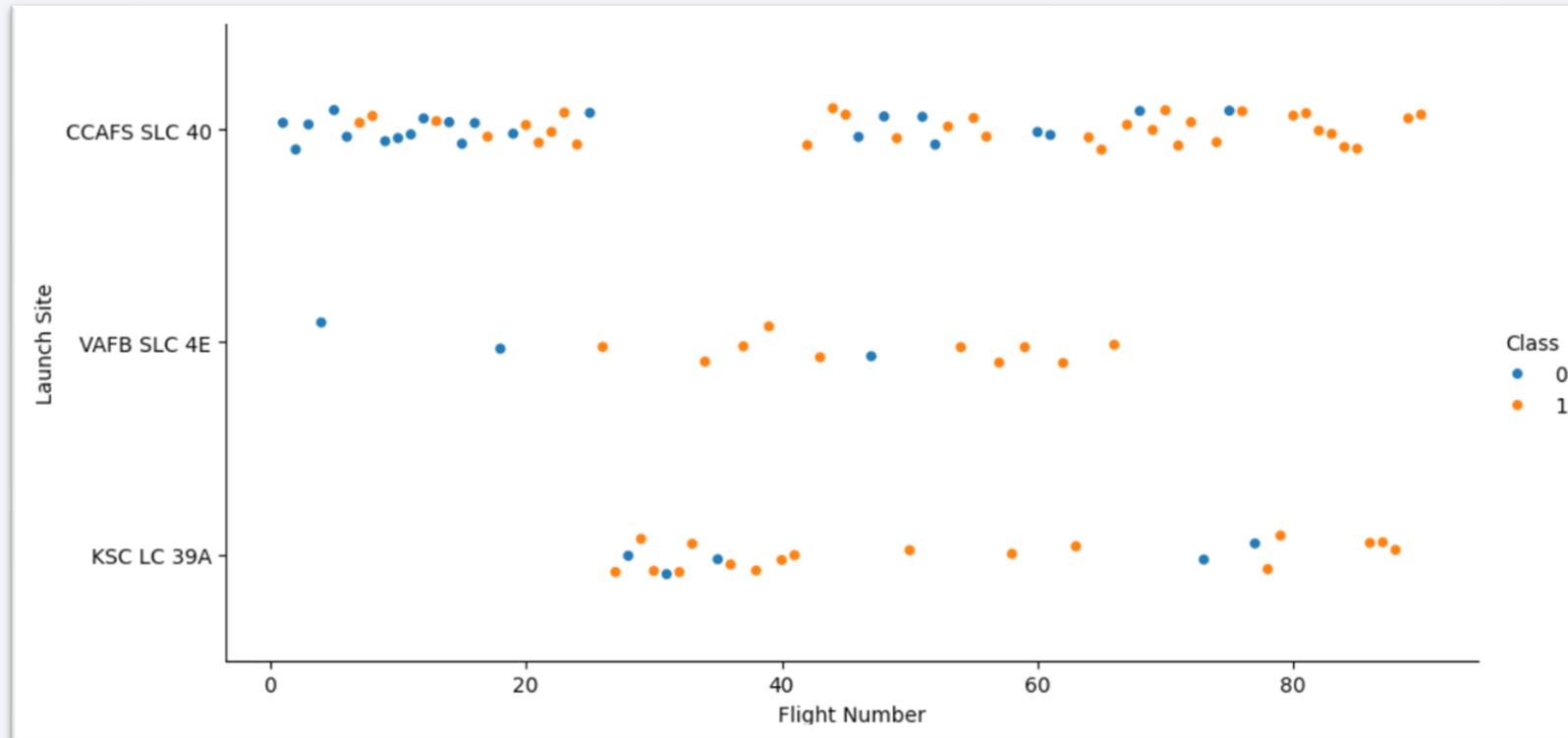
- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

The background of the slide is an abstract composition. It features a dark blue gradient on the left side, which transitions into a complex pattern of diagonal streaks and lines in shades of blue, red, and teal on the right. These streaks have a textured, almost woven appearance, suggesting a digital or data-driven theme. The overall effect is dynamic and modern.

Section 2

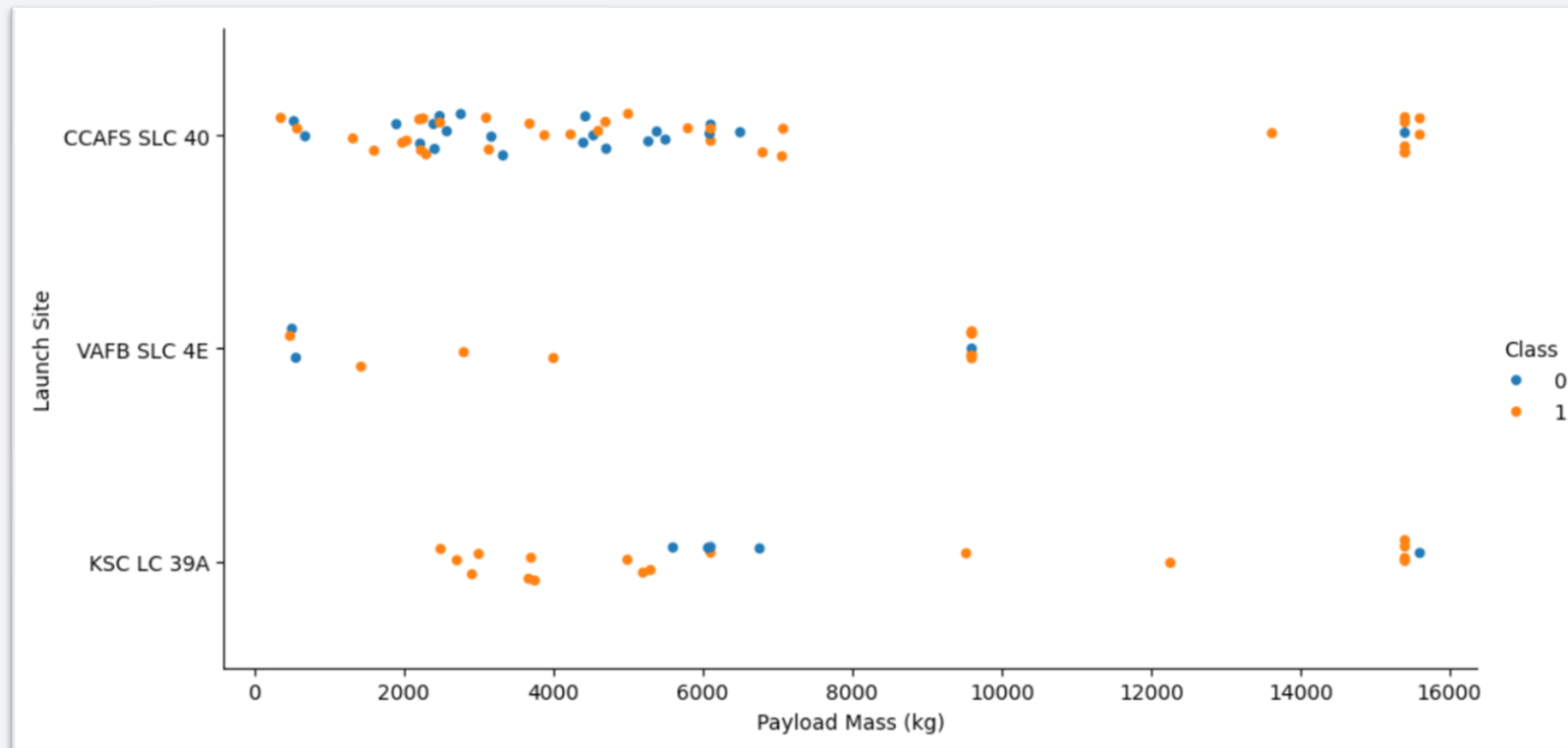
Insights drawn from EDA

Flight Number vs. Launch Site



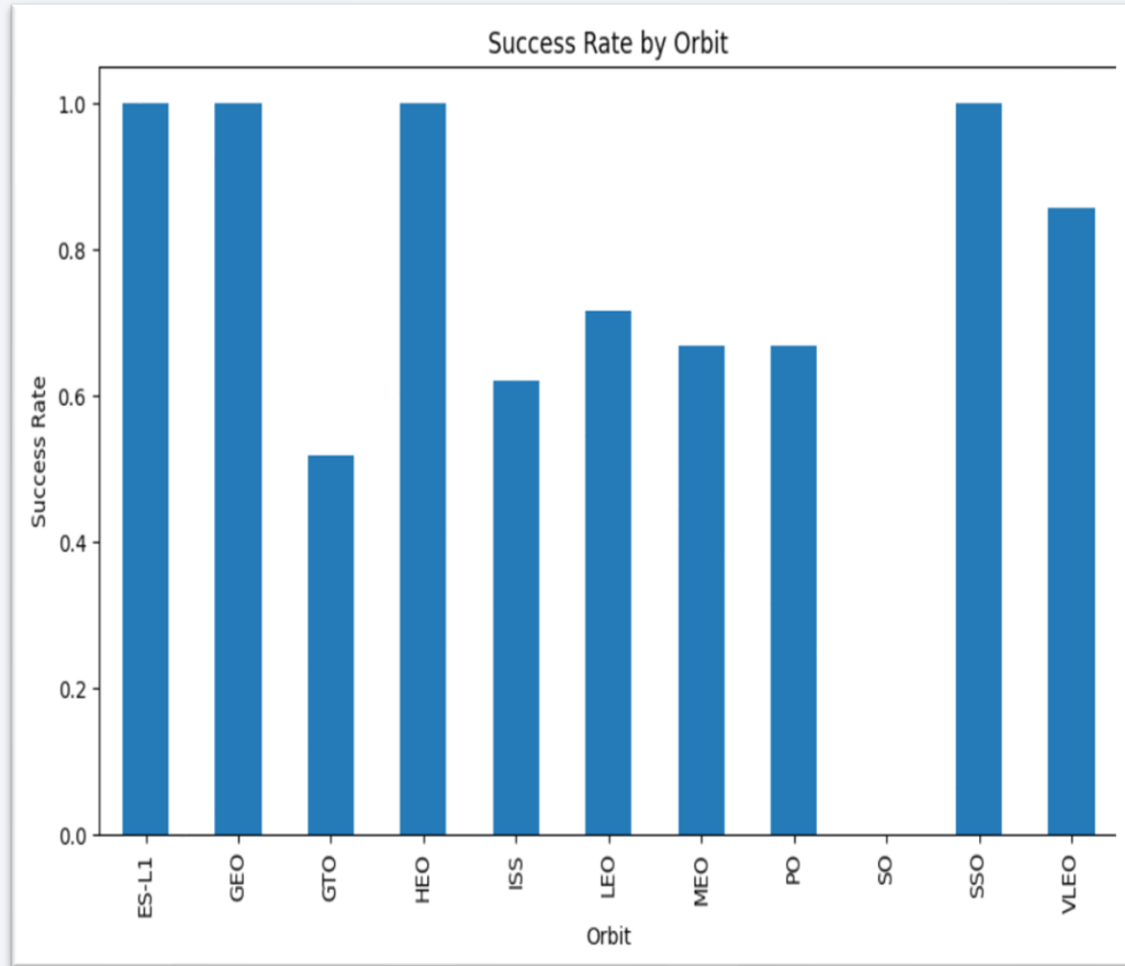
Launch success rates (Class 1) increase significantly with higher flight numbers as experience grew. Early flights at CCAFS SLC 40 had more failures, while later launches across all sites achieved near-consistent success.

Payload vs. Launch Site



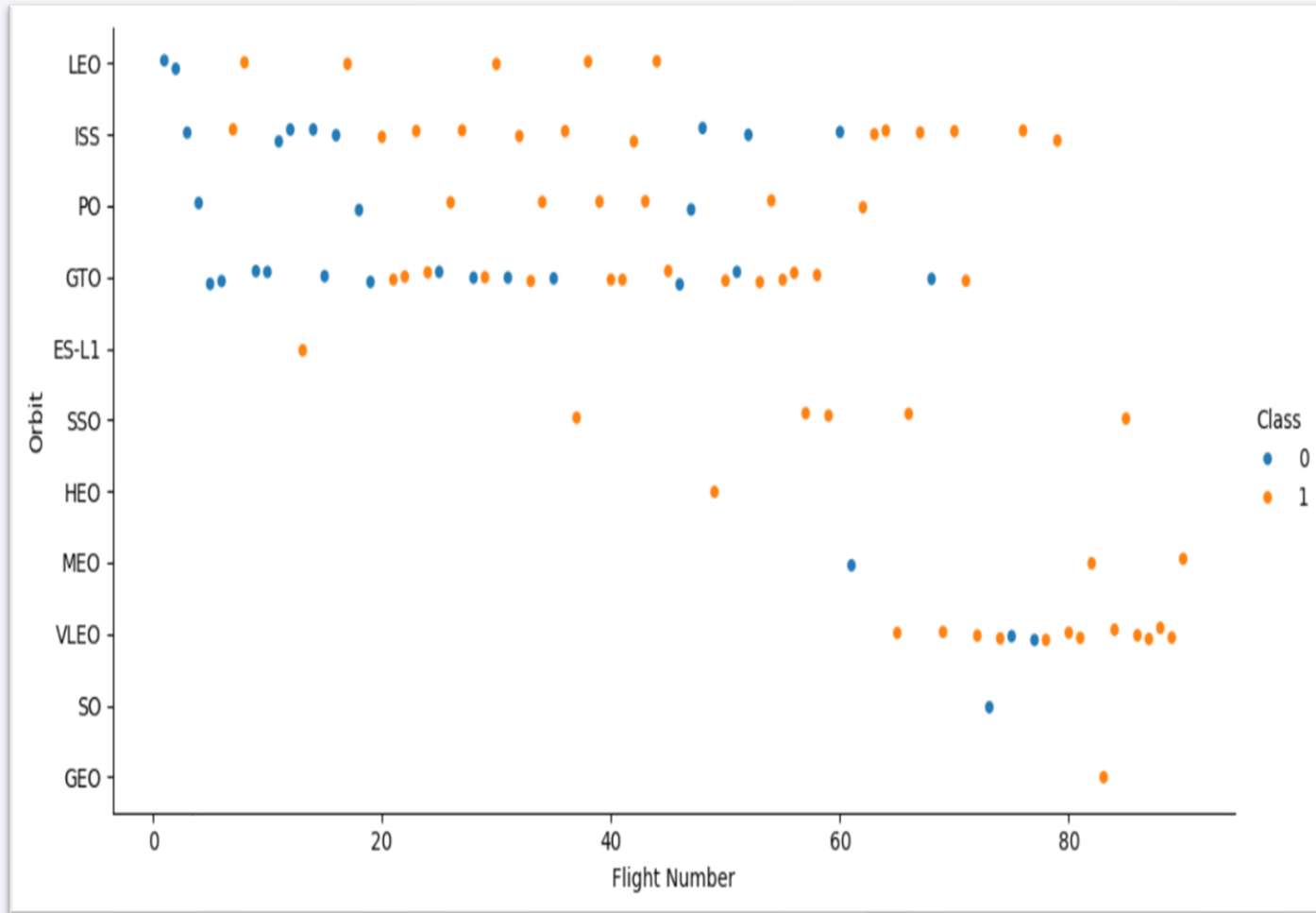
Missions with heavy payloads (greater than 10,000 kg) show a very high success rate across all launch sites. In contrast, most failures occurred with lighter payloads (below 7,000 kg), primarily at CCAFS SLC 40.

Success Rate vs. Orbit Type



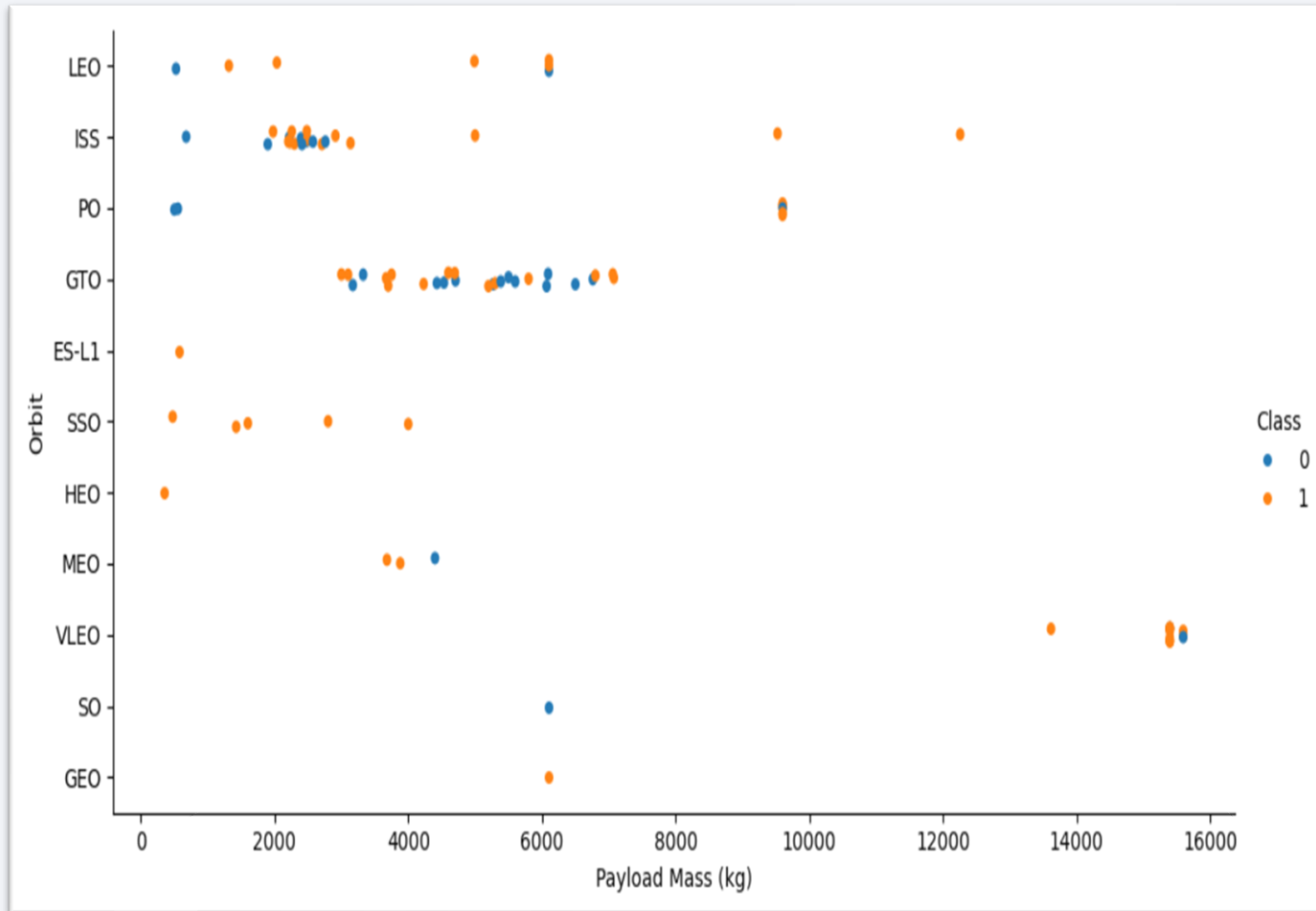
- Highest success rate (100%): Orbits **ES-L1**, **GEO**, **HEO**, and **SSO** achieved a perfect 1.0 success rate, followed closely by **VLEO** (around 85%).
- Moderate success rate (50%–75%): **LEO**, **MEO**, **PO**, **ISS**, and **GTO** show moderate landing success rates ranging between roughly 50% and 72%.
- Lowest success rate (0%): **SO** has a 0% success rate, indicating no successful landings for this orbit type.

Flight Number vs. Orbit Type



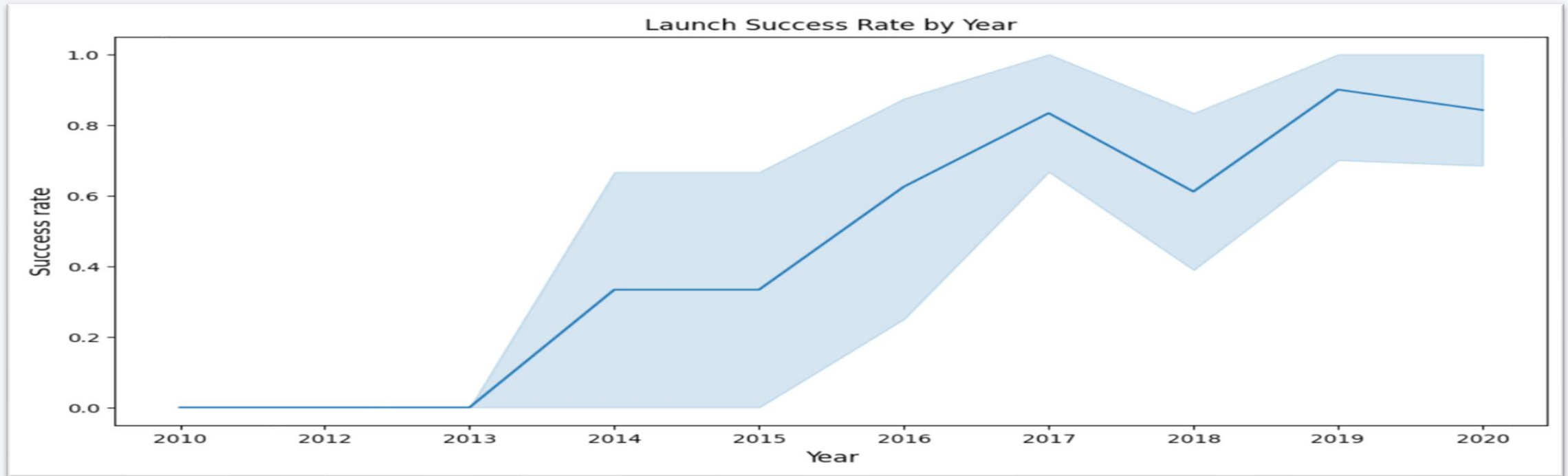
- Early flights mainly targeted **LEO**, **ISS**, and **GTO** with mixed landing outcomes, but landing success rates (Class 1) improved significantly across these orbits as flight numbers increased.
- Later missions (higher flight numbers) expanded into orbits like **VLEO** and **SSO**, demonstrating near-perfect landing success due to matured recovery technology.

Payload vs. Orbit Type



- Heavy payloads (greater than 10,000 kg) are exclusively carried to **VLEO** and **ISS**, with **VLEO** showing a near-perfect landing success rate.
- Orbits with light-to-medium payloads (below 7,000 kg) such as **GTO**, **PO**, and **ISS** exhibit mixed outcomes, where almost all landing failures occurred.

Launch Success Yearly Trend



- The launch success rate remained at 0% from 2010 to 2013, followed by a steady upward trend starting in 2014.
- Overall success peaked near 90% by 2019, reflecting significant technological maturity and reliability in booster landings over time.

All Launch Site Names

```
%sql SELECT DISTINCT Launch_Site FROM SPACEXTABLE;
```

The query retrieves all four distinct launch sites used in SpaceX missions: CCAFS LC-40, VAFB SLC-4E, KSC LC-39A, and CCAFS SLC-40.

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Launch Site Names Begin with 'CCA'

```
%sql SELECT * FROM SPACEXTABLE WHERE Launch_Site LIKE 'CCA%' LIMIT 5;
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

The query filters and displays the **first five** mission records launched from Cape Canaveral sites starting with "CCA" (CCAFS LC-40), all of which were early Falcon 9 flights carrying Dragon payloads to LEO.

Total Payload Mass

```
%sql SELECT SUM(PAYLOAD_MASS_KG_) AS Total_Payload_Mass_KG FROM SPACEXTABLE WHERE Customer = 'NASA (CRS)';
```

Total_Payload_Mass_KG

45596

The query calculates the total payload mass carried for NASA's Commercial Resupply Services (CRS) missions, which amounts to **45,596 kg**.

Average Payload Mass by F9 v1.1

```
%sql SELECT AVG(PAYLOAD_MASS_KG_) AS Average_Payload_Mass_KG FROM SPACEXTABLE WHERE Booster_Version = 'F9 v1.1';
```

Average_Payload_Mass_KG

2928.4

The query calculates the average payload mass carried specifically by Falcon 9 v1.1 boosters, resulting in 2,928.4 kg.

First Successful Ground Landing Date

```
-- Task 3: Find the date of the first successful landing on a ground pad
%sql SELECT MIN(Date) AS First_Successful_Ground_Pad_Landing FROM SPACEXTABLE WHERE Landing_Outcome = 'Success (ground pad)';
```

First_Successful_Ground_Pad_Landing

2015-12-22

The query retrieves the date of SpaceX's first successful ground pad landing, which occurred on **December 22, 2015**.

Successful Drone Ship Landing with Payload between 4000 and 6000

```
%sql SELECT Booster_Version, PAYLOAD_MASS_KG_ FROM SPACEXTABLE WHERE Landing_Outcome = 'Success (drone ship)' AND PAYLOAD_MASS_KG_ > 4000 AND PAYLOAD_MASS_KG_ < 6000;
```

Booster_Version	PAYLOAD_MASS_KG_
F9 FT B1022	4696
F9 FT B1026	4600
F9 FT B1021.2	5300
F9 FT B1031.2	5200

The query lists four Falcon 9 Full Thrust (F9 FT) boosters that achieved successful drone ship landings while carrying payloads between 4,000 kg and 6,000 kg.

Total Number of Successful and Failure Mission Outcomes

```
%sql SELECT Mission_Outcome, COUNT(Mission_Outcome) AS Total_Missions FROM SPACEXTABLE GROUP BY Mission_Outcome;
```

Mission_Outcome	Total_Missions
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

The query groups mission outcomes, showing 100 total successful launches (with slight label variations) compared to only 1 in-flight failure.

Boosters Carried Maximum Payload

```
%sql SELECT DISTINCT Booster_Version, PAYLOAD_MASS_KG_ AS Maximum_Payload_Mass_KG FROM SPACEXTABLE WHERE PAYLOAD_MASS_KG_ = (SELECT MAX(PAYLOAD_MASS_KG_) FROM SPACEXTABLE);
```

The query identifies all Falcon 9 Block 5 (F9 B5) booster versions that carried the maximum recorded payload mass of 15,600 kg.

Booster_Version	Maximum_Payload_Mass_KG
F9 B5 B1048.4	15600
F9 B5 B1049.4	15600
F9 B5 B1051.3	15600
F9 B5 B1056.4	15600
F9 B5 B1048.5	15600
F9 B5 B1051.4	15600
F9 B5 B1049.5	15600
F9 B5 B1060.2	15600
F9 B5 B1058.3	15600
F9 B5 B1051.6	15600
F9 B5 B1060.3	15600
F9 B5 B1049.7	15600

2015 Launch Records

```
%%sql
SELECT
    CASE substr(Date, 6, 2)
        WHEN '01' THEN 'January'
        WHEN '02' THEN 'February'
        WHEN '03' THEN 'March'
        WHEN '04' THEN 'April'
        WHEN '05' THEN 'May'
        WHEN '06' THEN 'June'
        WHEN '07' THEN 'July'
        WHEN '08' THEN 'August'
        WHEN '09' THEN 'September'
        WHEN '10' THEN 'October'
        WHEN '11' THEN 'November'
        WHEN '12' THEN 'December'
    END AS Month,
    Landing_Outcome,
    Booster_Version,
    Launch_Site
FROM SPACEXTABLE
WHERE Landing_Outcome = 'Failure (drone ship)'
    AND substr(Date, 1, 4) = '2015';
```

Month	Landing_Outcome	Booster_Version	Launch_Site
January	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
April	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

The query retrieves drone ship landing failures in 2015, showing two failed attempts in January and April using Falcon 9 v1.1 boosters launched from CCAFS LC-40.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

```
%%sql
SELECT Landing_Outcome, COUNT(Landing_Outcome) AS Outcome_Count
FROM SPACEXTABLE
WHERE Date BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY Landing_Outcome
ORDER BY Outcome_Count DESC;
```

The query ranks landing outcomes from June 2010 to March 2017, showing "No attempt" as the most frequent outcome (10), followed by 5 successful and 5 failed drone ship landings.

Landing_Outcome	Outcome_Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

A satellite view of Earth from space, showing the curvature of the planet and the glowing lights of cities at night. The background is a deep blue gradient.

Section 3

Launch Sites Proximities Analysis

Global Distribution of Launch Sites



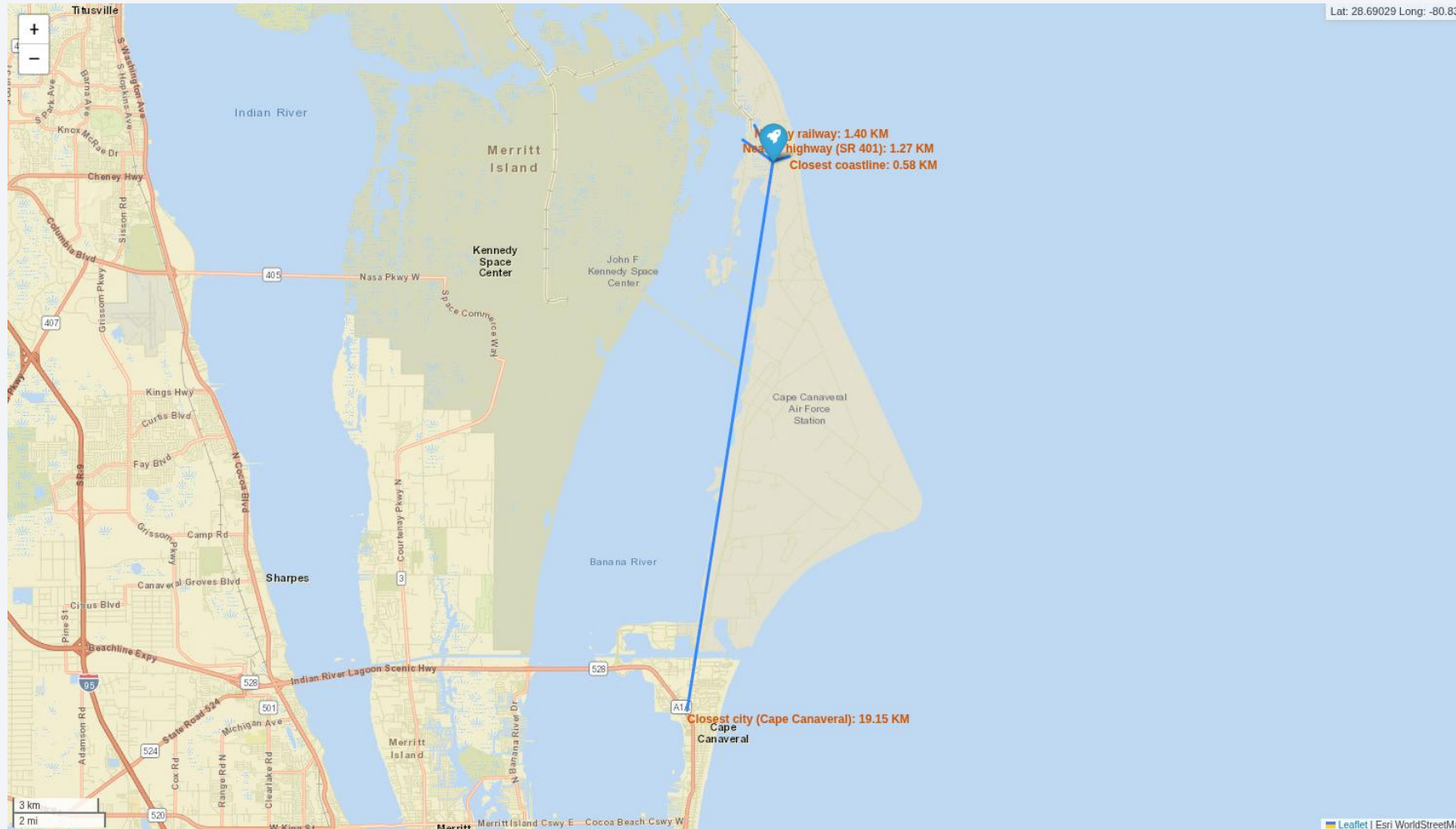
This map shows the global distribution and geographic locations of all launch sites.

Launch Outcomes: Success vs Failure

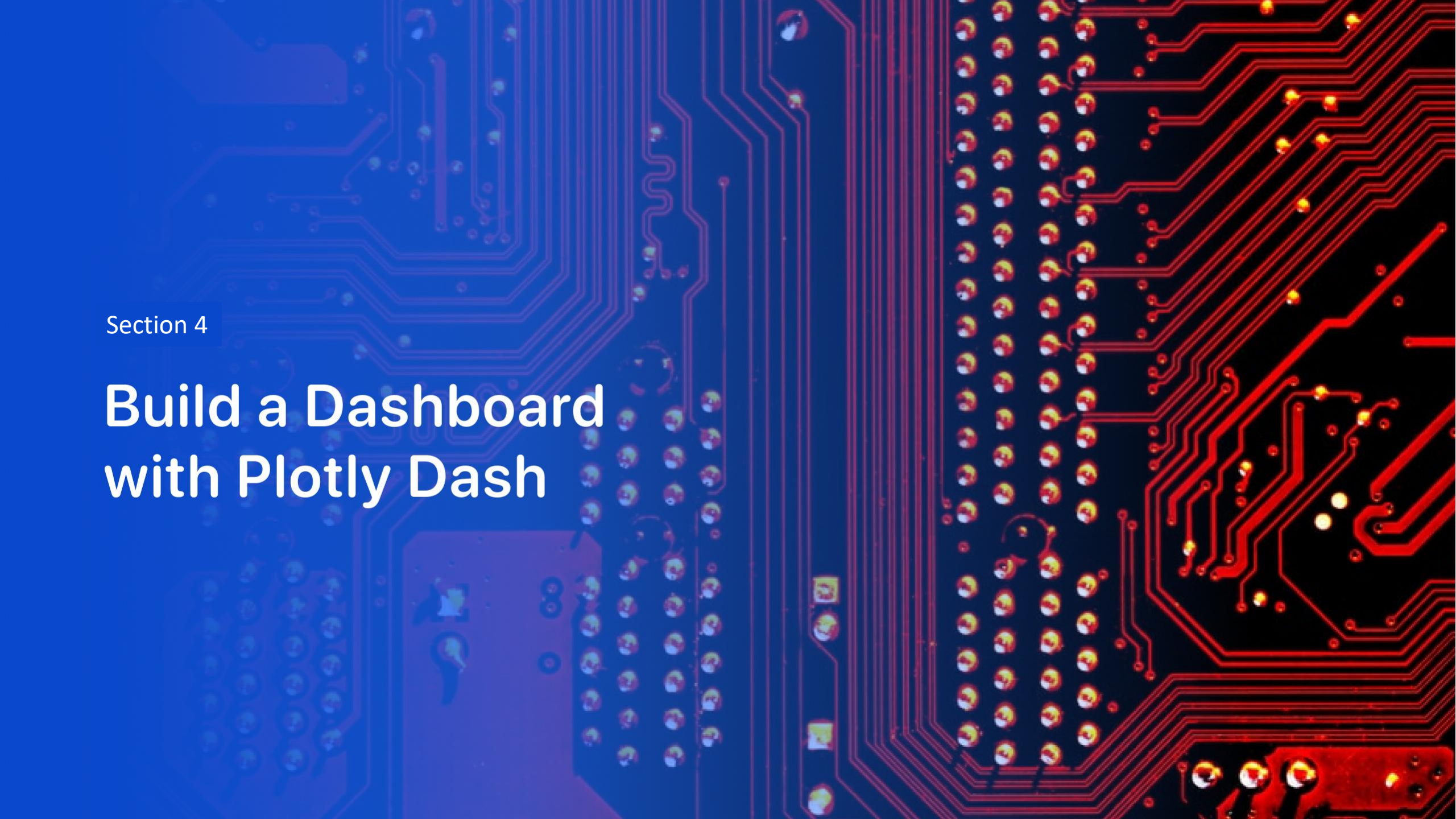


This map shows the launch outcomes using green markers for successful launches and red markers for failed launches.

CCAFS SLC-40 Proximity Analysis



This map shows the proximity of the selected CCAFS SLC-40 launch site to the coastline, railway, highway, and nearby city, with the distances displayed in kilometers.



Section 4

Build a Dashboard with Plotly Dash

Total Success Launches By Site

Total Success Launches By Site



The pie chart illustrates the distribution of total successful launches across four sites, with **KSC LC-39A** leading as the primary contributor at **41.7%**. In contrast, **CAFS SLC-40** recorded the smallest share of successful launches at **12.5%**.

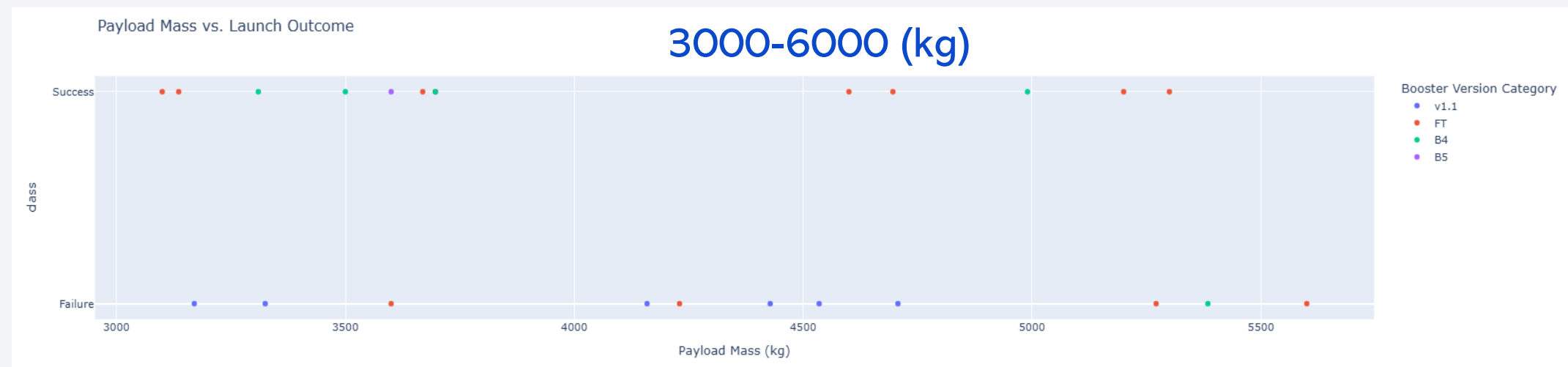
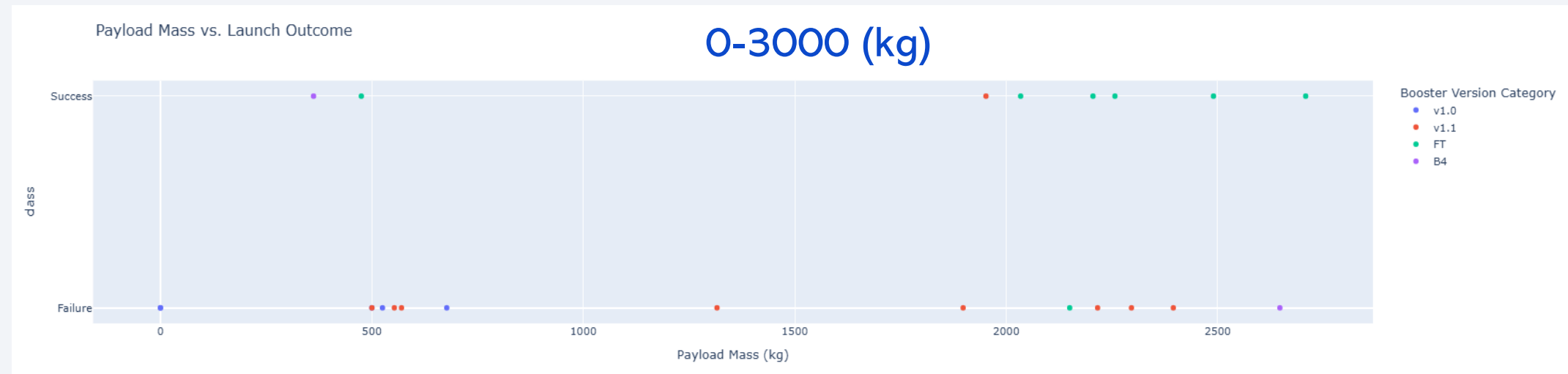
Highest Launch Success (KSC LC-39A)

Success vs. Failure for site KSC LC-39A

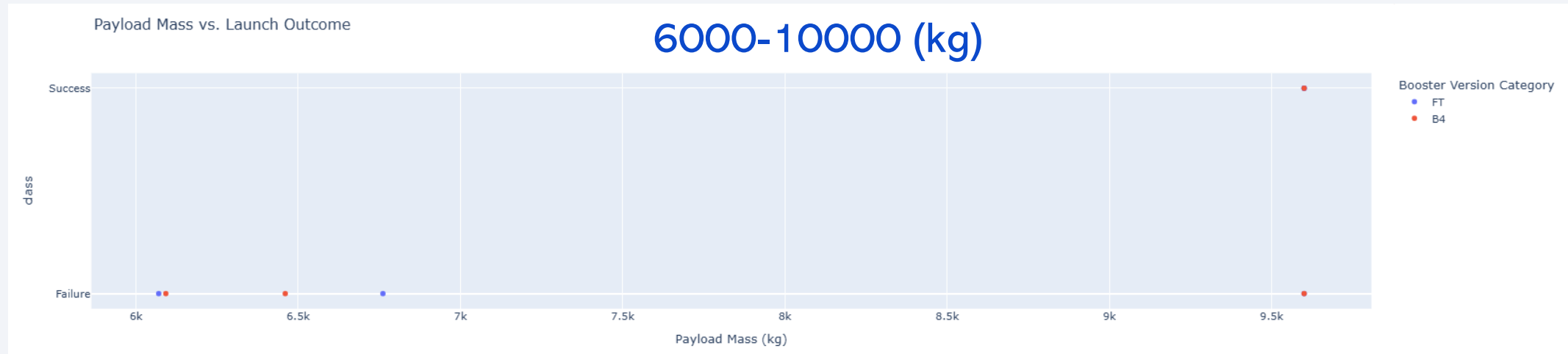


The pie chart shows that launch site **KSC LC-39A** achieved a high success rate of **76.9%**, with failures accounting for only **23.1%** of its total missions.

Payload vs Launch Outcome



Payload vs Launch Outcome

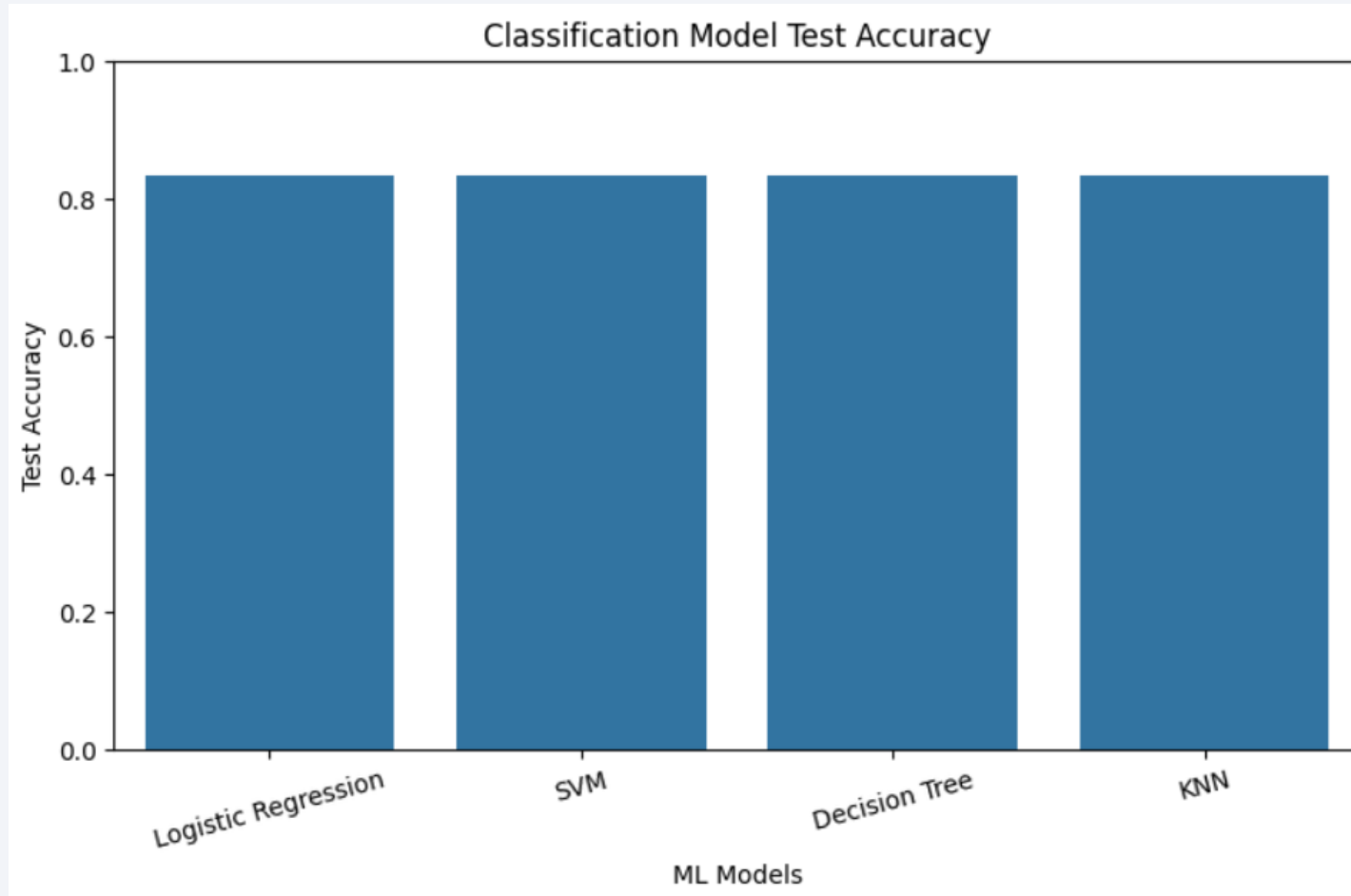


- Higher payload ranges achieve better success: Missions carrying payloads between 2,000–4,000 kg and above 5,000 kg demonstrate noticeably higher success rates compared to low-payload missions (under 2,000 kg), where most early failures occurred.
- Modern boosters deliver superior reliability: Advanced booster versions (FT, B4, and B5) dominate successful landings across all mass ranges, whereas early versions (v1.0 and v1.1) account for the majority of landing failures.

Section 5

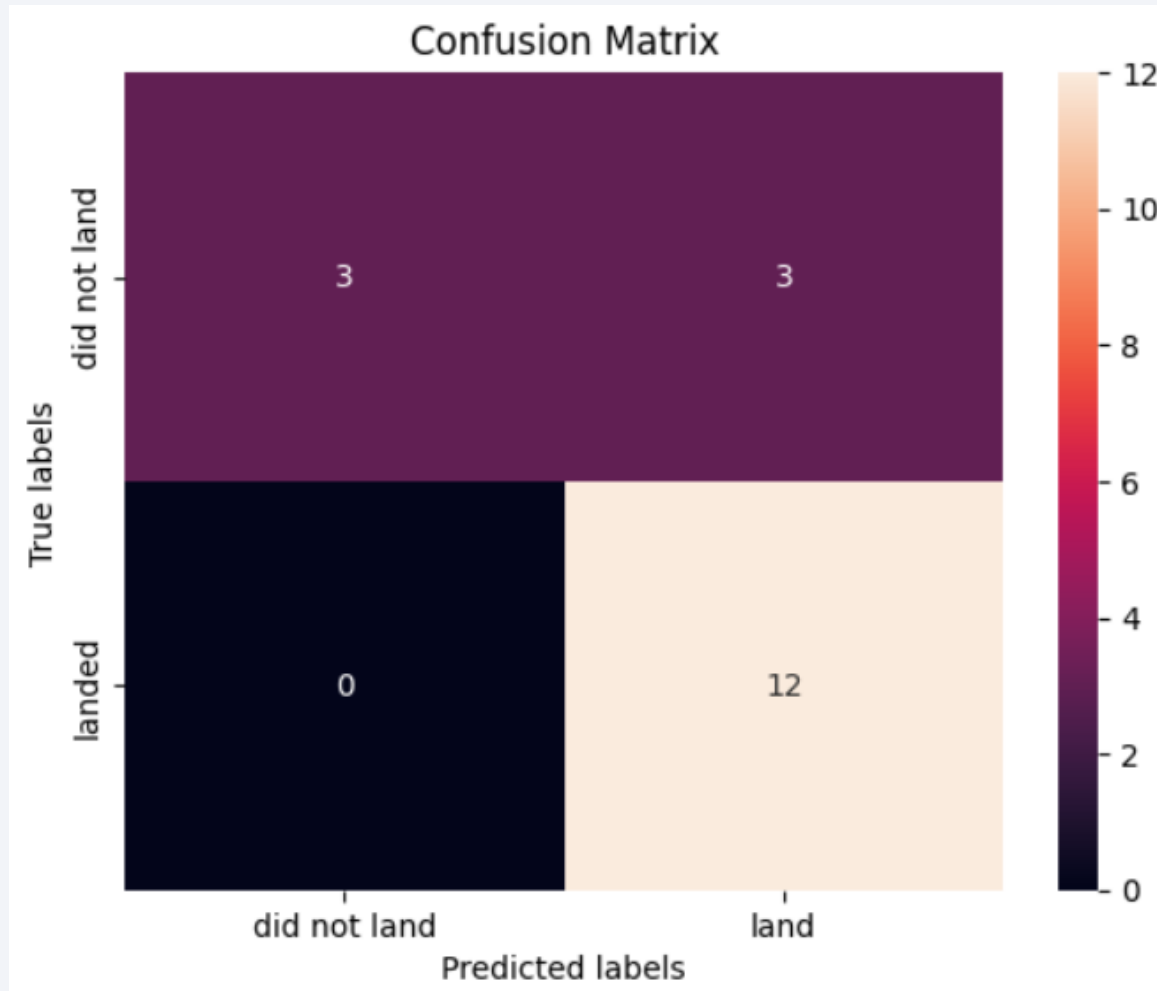
Predictive Analysis (Classification)

Classification Accuracy



- Test Performance: All models achieved 83.33% test accuracy.
- Cross-Validation: Decision Tree scored highest at 88.9%, while others achieved 83.3%.
- Recommendation: Logistic Regression or SVM for consistent performance without overfitting.

Confusion Matrix



- Confusion Matrix: The model correctly predicted 12 successful landings and 3 failures, with 3 false positives and 0 false negatives.
- Identical Results: All four algorithms produced the same confusion matrix due to the 18-sample test set having strong feature separation.

Conclusions

- Significant Success Growth: Landing success increased steadily from 0% (2010–2013) to nearly 90% by 2019 due to technological maturity and booster iterations.
- Site & Payload Influence: KSC LC-39A achieved the highest overall success rate (76.9%), while missions carrying heavy payloads (>10,000 kg) had near-perfect landing success.
- Booster Performance: Modern booster variants (FT, B4, and B5) dramatically outperformed early iterations (v1.0 and v1.1), driving the vast majority of successful recoveries.
- Accurate Predictive Modeling: All evaluated classification models achieved 83.33% test accuracy, proving the practical feasibility of predicting first-stage reusability to forecast launch costs.

Appendix

- Project Repository: Jupyter Notebooks on GitHub:
<https://github.com/johnsonteks/Data-Science-Capstone>
- Core Technologies & Libraries: Python (Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, Folium, Plotly Dash), SQLite/SQL, and BeautifulSoup.
- Data Sources: SpaceX REST API and Wikipedia Falcon 9 historical launch logs.

Thank you!

